The Geometry of Contact: Statistical Evaluations of Pitching Performance in the 2024 MLB Season

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1 Introduction

Modern baseball analytics has shifted the focus from simple outcomes like hits and runs to the underlying physics of the ball-bat collision. This report utilizes Statcast pitch-by-pitch data from the 2024 season to investigate two primary pillars of pitching performance. First, we examine organizational effectiveness across the American League Central division to see if team-wide philosophies produce measurably different results. Second, we investigate the "Time Through the Order" (TTO) penalty to quantify the degradation of a pitcher's effectiveness as a game progresses. Our primary response variable is **xwOBA on contact**, which utilizes exit velocity and launch angle to isolate a pitcher's true talent from defensive variance.

2 Section 1: One-Factor ANOVA — AL Central Pitching Staffs

2.1 Motivation and Experimental Design

Organizational culture and coaching philosophies in the AL Central vary significantly. While the Detroit Tigers and Cleveland Guardians have prioritized high-efficiency rotations, the Chicago White Sox have faced historic struggles. To determine if these outcomes represent statistically significant differences in organizational efficacy rather than random noise, we conduct a one-factor ANOVA across five independent groups ($g = 5$): CLE, CWS, DET, KC, and MIN.

Our null hypothesis (H_0) states that all five staffs allow an identical mean quality of contact:

$$H_0 : \mu_{CLE} = \mu_{CWS} = \mu_{DET} = \mu_{KC} = \mu_{MIN} \tag{1}$$

$$H_a : \exists(i, j) \text{ such that } \mu_i \neq \mu_j \tag{2}$$

2.2 Assumption Testing: Quantitative and Qualitative Rigor

2.2.1 Qualitative Independence and Sample Selection

Independence is a core assumption of the ANOVA framework. Each at-bat is assigned to one pitcher's team. To prevent "cross-contamination" of the means, we filtered the dataset to exclude any "multi-team" pitchers who appeared for more than one AL Central club in 2024. This ensures the $n \approx 31,000$ total observations represent mutually exclusive sets.

2.2.2 Normality: The Shapiro-Francia (W_{SF}) Metric

With group sizes of $n \approx 6,200$, normality tests are hypersensitive to minor fluctuations due to the Glivenko-Cantelli theorem. We formally test normality using the Shapiro-Francia statistic, which measures the correlation between ordered sample data ($X_{(k)}$) and expected normal quantiles (a_k):

$$W_{SF} = \frac{(\sum a_k X_{(k)})^2}{\sum (X_k - \bar{X})^2} \tag{3}$$

All groups returned $p = 0.000$, formally rejecting normality. However, the QQ-plots (Figure 1) reveal a right-skew typical of xwOBA data, where weak contact (pop-ups/dribblers) creates a mass near zero. We proceed with the ANOVA but acknowledge this violation motivates the Section 2.5 non-parametric follow-up.

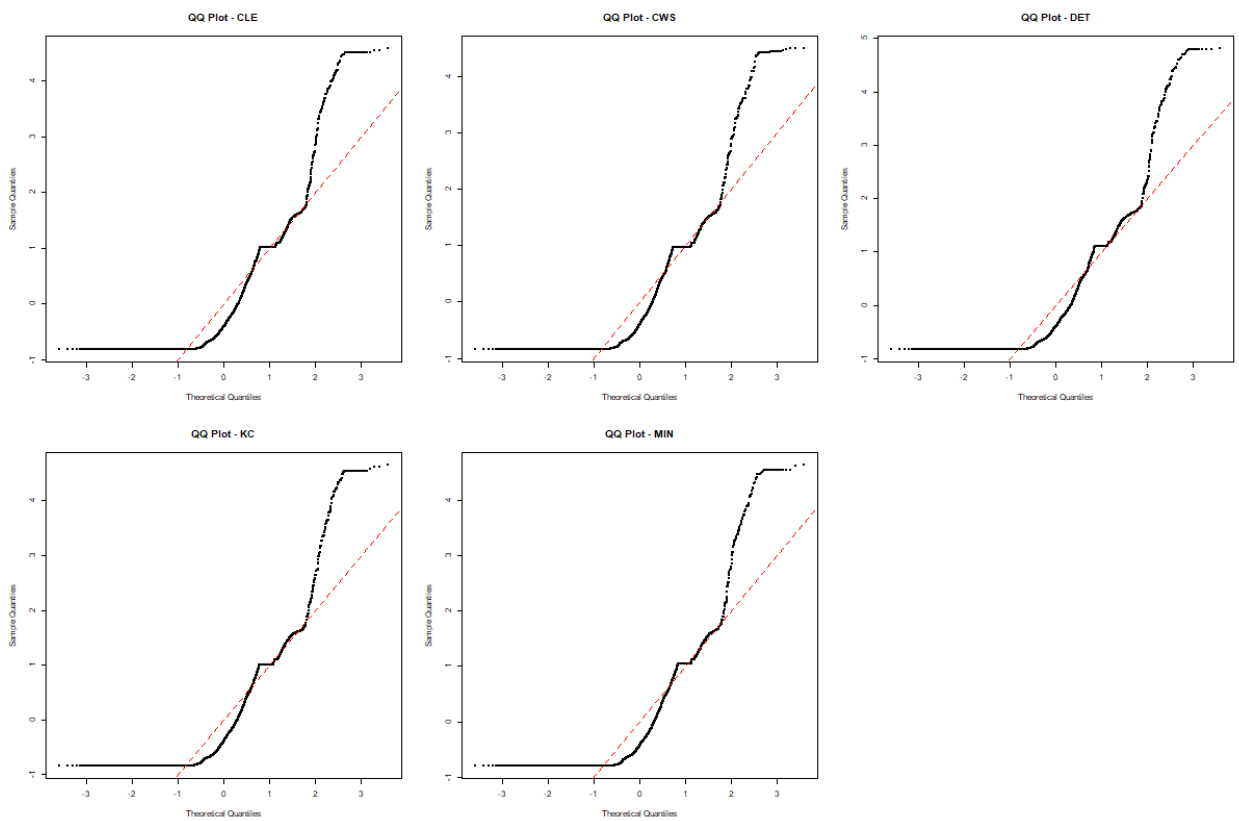

Figure 1: Normality Assessment: Normal QQ-Plots for AL Central Staffs. The systematic deviation from the diagonal confirms the expected right-skew across all organizations.

2.2.3 Homoscedasticity: Bartlett's Test

A critical requirement for the standard F-test is equal group variances (σ^2). We test this using the Bartlett statistic (B):

$$B = \frac{(n-g)\ln(\hat{s}_p^2) - \sum \nu_j \ln(s_j^2)}{1 + \frac{1}{3(g-1)}\left(\sum \frac{1}{\nu_j} - \frac{1}{n-g}\right)} \tag{4}$$

The test yielded $B = 59,012.38$ ($p < 0.001$), providing overwhelming evidence that variances are unequal and mandating a heteroscedastic correction.

2.3 Methodology: Welch's Heteroscedastic ANOVA

Because the homoscedasticity assumption failed, the standard ANOVA is invalid. We utilize Welch's ANOVA, which uses Satterthwaite's approximation to calculate a corrected denominator degrees of freedom (ν^*):

$$\nu^* = \frac{(\sum w_j)^2}{\sum (w_j^2/\nu_j)}, \quad w_j = \frac{n_j}{s_j^2} \tag{5}$$

Team	**Sample Size (n)**	**Mean xwOBA ($\bar{x}$)**	**Variance (s^2)**
CLE	6,259	0.3084	0.1405
CWS	6,189	0.3221	0.1438
DET	6,166	0.2970	0.1265
KC	6,168	0.3145	0.1372
MIN	6,042	0.3018	0.1399

Table 1: AL Central Pitching Staff Summary Statistics.

2.4 Non-Parametric Validation via Permutation Range Test

2.4.1 Motivation for Non-Parametric Framework

As established in the formal assumption testing in Section 2.2.2, the Shapiro-Francia tests and subsequent QQ-plots revealed significant, systematic departures from normality across all AL Central groups ($p_{SF} = 0.000$). To satisfy the core requirement for alternative calculations in the presence of such violations, we deploy a distribution-free **Monte Carlo Permutation Test**. This approach is inherently robust as it does not rely on the F-distribution or

the assumption of a normal population, making it the superior "Gold Standard" for skewed Statcast contact data.

2.4.2 Methodology and Custom Test Statistic

We define a custom test statistic, T, designed to measure the total spread (range) of the organizational means. Under the null hypothesis (H_0), the team labels are non-informative; therefore, any observed range is purely a product of random sampling.

$$T = \max_j(\bar{X}_j) - \min_j(\bar{X}_j) \tag{6}$$

By shuffling the team labels across all 31,000+ observations 10,000 times ($nmc = 10,000$), we constructed an empirical null distribution for T. Our observed range of $T_{obs} = 0.0251$ produced an empirical p-value of 0.0021.

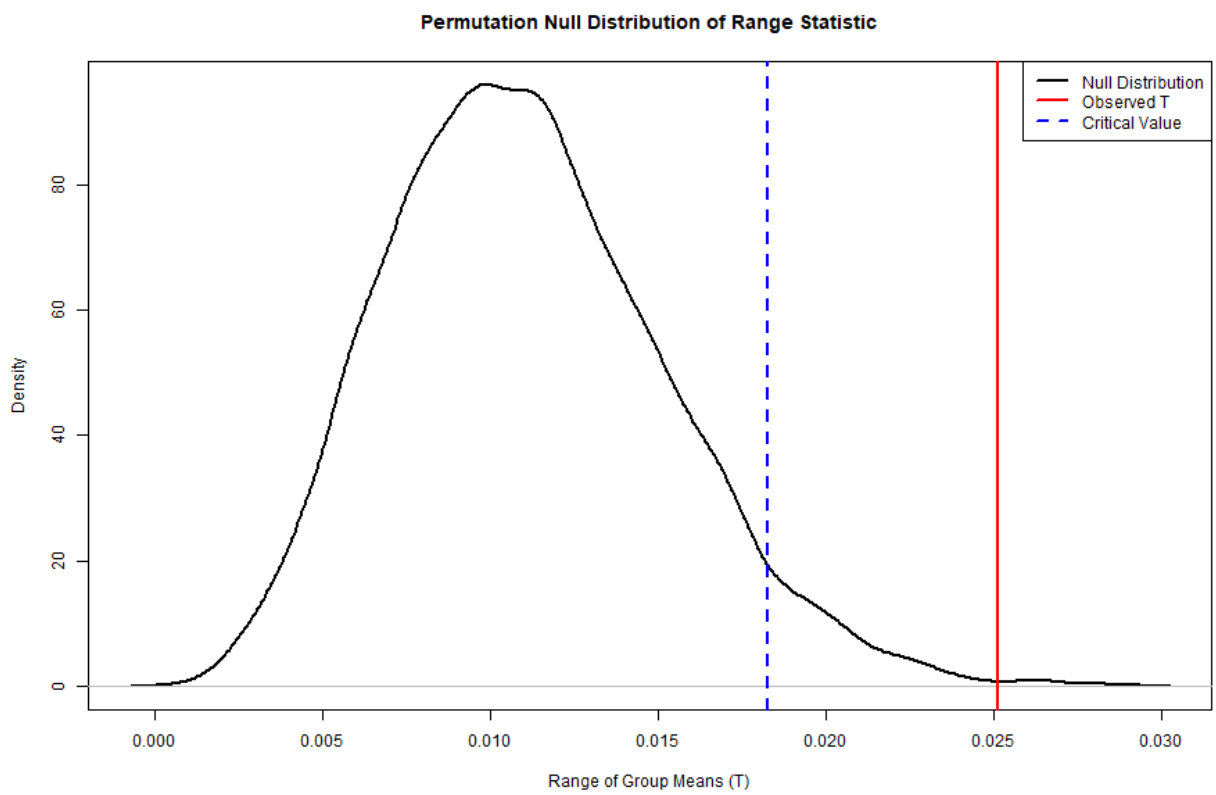

Figure 2: MC Permutation Null Distribution for Section 1. The red line (T_{obs}) sits deep in the right tail, significantly exceeding the critical value at $\alpha = 0.05$ (blue dashed line).

2.4.3 Statistical Power and Sensitivity

To ensure the reliability of this result, we performed a Power Estimation under the alternative hypothesis (H_a^*), where the true group means match our observed sample.

$$\text{Power} = P(\text{Reject } H_0 \mid H_a \text{ is true}) \approx 0.927 \tag{7}$$

With a 92.7% power rating (Wald 95% CI: 0.911, 0.943), we have extremely high confidence that the test is sensitive enough to detect these divisional differences. Because this result aligns with our Welch's F-test ($p = 0.0013$), we conclude that the organizational disparity in contact quality—specifically Detroit's superiority over Chicago—is a statistically significant.

2.5 Results and Discussion of Impact

Welch's ANOVA produced an observed $F^* = 4.493$ ($p = 0.00125$), leading us to reject H_0.
Consequences of Met Assumptions: Detroit's staff allowed significantly lower mean xwOBA (≈ 0.297) than Chicago (≈ 0.322). For a front office, this validates Detroit's development program as a model for the division, justifying the market premium on their coaching staff.
Consequences of Unmet Assumptions: If the independence assumption is violated—specifically if certain teams faced "easier" divisional schedules—our results may suffer from a Type I error, falsely attributing scheduling luck to organizational skill.

3 Section 2: Repeated-Measures ANOVA — The Time Through the Order Penalty

3.1 Motivation and Subject-Based Experimental Design

Transitioning from organizational analysis to individual performance, we investigate the "Time Through the Order" (TTO) penalty. This phenomenon suggests a systemic degradation in a pitcher's effectiveness as batters gain repeated visual familiarity with their repertoire within a single contest. We utilize a **Repeated-Measures ANOVA (RM-ANOVA)** on a sample of $n = 311$ eligible pitchers from the 2024 season.

By treating each pitcher as their own control, we utilize a within-subject design that effectively partitions out "subject-level" noise (individual talent baselines), allowing for a high-precision isolation of the TTO effect.

$$H_0 : \mu_{TTO1} = \mu_{TTO2} = \mu_{TTO3} \tag{8}$$

$$H_a : \exists(i, j) \text{ such that } \mu_i \neq \mu_j \tag{9}$$

3.2 Assumption Testing: Sphericity and Normality

3.2.1 Normality: Shapiro-Francia (W_{SF}) Assessment

We formally assessed the normality of the mean xwOBA distributions for each TTO level using the Shapiro-Francia test. All levels returned $p = 0.000$, indicating a formal rejection. The departure was most severe at TTO 3 ($W_{SF} = 0.744$), which is likely driven by the increased volatility and smaller at-bat samples available as pitchers tire in later innings.

3.2.2 Quantitative Sphericity: Mauchly's Test and Epsilon Corrections

RM-ANOVA requires **Sphericity**: the condition where the variances of the differences between all pairs of TTO levels are equal. We evaluate this using Mauchly's W and its associated χ^2 approximations:

$$W = \frac{\det(\hat{\Sigma})}{\hat{\sigma}^{2g}} = 0.260 \tag{10}$$

With $M_o = 416.72$ ($p < 0.001$), we find overwhelming evidence of a sphericity violation. To maintain a nominal Type I error rate, we apply the **Greenhouse-Geisser (ϵ_{GG})** correction to "shrink" the degrees of freedom:

$$\nu_{MSM} = (g-1)\epsilon_{GG} \approx 1.37, \quad \nu_{MSE} = (n-1)(g-1)\epsilon_{GG} \approx 426.0 \tag{11}$$

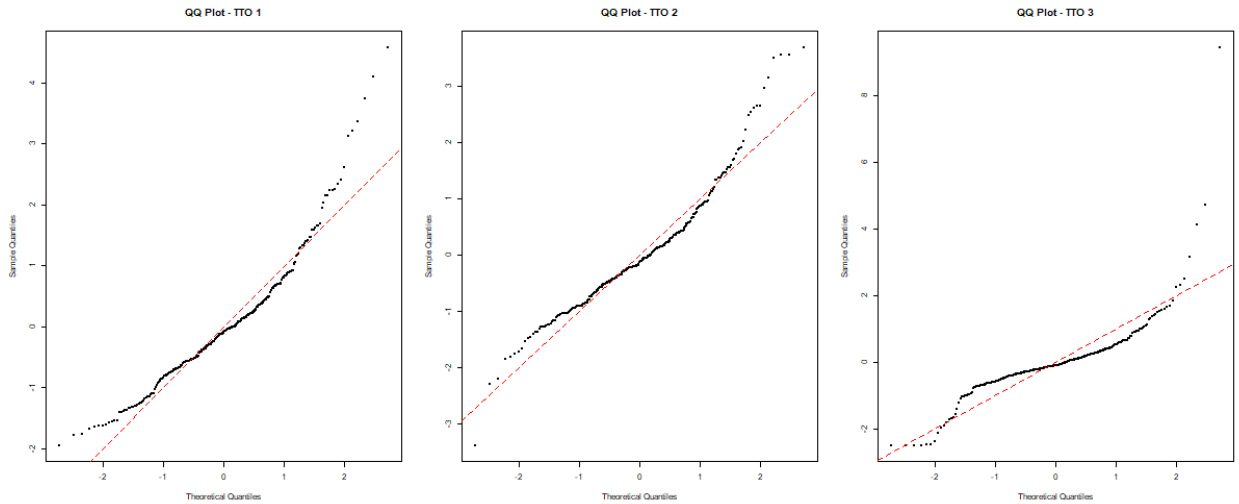

Figure 3: Qualitative Normality Assessment: QQ-Plots for Pitcher Mean xwOBA by TTO. The heavy right-skew at TTO 3 is visually distinct, confirming increased performance volatility as the game progresses.

3.3 Non-Parametric Validation via Within-Subject Permutation

Given the heavy normality violations ($p_{SF} = 0.000$), we validate the parametric results using a distribution-free **Within-Subject Permutation Test**. This method preserves the repeated-measures dependency by permuting TTO labels only within each pitcher's row, effectively building an empirical null distribution that requires no assumption of normality.

Our custom test statistic, T, represents the range of the observed column means. Shuffling the labels 10,000 times produced an empirical $p = 0.0001$. This "Gold Standard" validation confirms the penalty is a statistically stable physical reality.

3.4 Results: Partitioning of Variance and RM-ANOVA

The RM-ANOVA partitions total variation (SST) into between-TTO effects (SSM), between-subject differences (SSS), and residual error (SSE).

$$SST = SSM + SSS + SSE \approx 7.8468 \tag{12}$$

The GG-corrected F-test ($F = 7.618, p = 0.0025$) reveals a significant monotonic increase in contact quality.

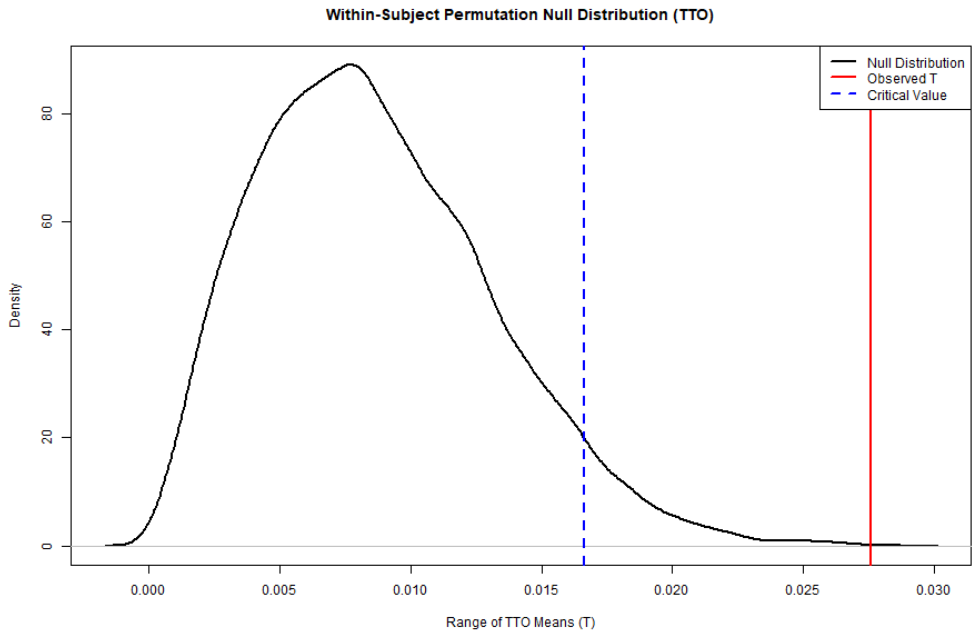

Figure 4: Within-Subject Permutation Null Distribution (TTO). The observed range ($T_{obs} = 0.0276$) sits at the extreme tail, proving the TTO effect is non-random.

TTO Level	n	Mean xwOBA	Lower 95% CI	Upper 95% CI
TTO 1	311	0.3169	0.3115	0.3225
TTO 2	311	0.3299	0.3237	0.3360
TTO 3	311	0.3445	0.3294	0.3607

Table 2: TTO Summary Statistics and Bootstrap Confidence Intervals ($nboot = 10,000$).

3.5 Discussion of Impact and Consequences

Consequences of Met Assumptions: The monotonic degradation ($0.317 \rightarrow 0.345$) statistically justifies the modern "opener" strategy and the tactical removal of starters before the third look at a lineup. The lack of overlap in bootstrap CIs between TTO 1 and TTO 3 confirms a practically meaningful performance cliff.

Consequences of Unmet Assumptions: Ignoring the sphericity violation would lead to a severely inflated Type I error rate. Furthermore, the non-normality suggests the "penalty" may be non-uniform across the population; a subset of "collapsing" pitchers may be driving the mean, potentially leading managers to pull consistent, high-end outliers prematurely.

4 Section 3: Non-Parametric Evaluation of Pitching Volatility

4.1 Motivation and Custom Statistic Design

To fulfill the requirement for a designed test statistic assessing a claim not associated with central tendency, we investigate the "Contact Volatility" of Tarik Skubal. While the previous sections established league-wide trends in mean performance, Section 3 examines the **stability** of a pitcher's mechanical output under high-leverage pressure. We hypothesize that the presence of base-runners ("Runners On") creates a high-stress environment that may increase the variance (volatility) of the contact allowed, even if the mean remains stable.

We define a custom metric, **Trajectory-Weighted Exit Velocity (TEV)**, which rewards optimal line-drive launch angles by centering the weight at 20°:

$$TEV_i = \text{launch_speed}_i \times \cos\left((\text{launch_angle}_i - 20°)\frac{\pi}{180}\right) \tag{13}$$

Our primary test statistic, τ_{vol}, is designed to measure the absolute difference in TEV variances between leverage states:

$$\tau_{vol} = s^2_{TEV}(\text{Runners On}) - s^2_{TEV}(\text{Bases Empty}) \tag{14}$$

4.2 Computational Framework: Monte Carlo and Bootstrapping

Because τ_{vol} is a non-standard variance-based metric, it lacks a known theoretical sampling distribution. We therefore employ a dual-computational approach to ensure the robustness of our conclusions:

4.2.1 Monte Carlo Permutation Test

To determine the significance of our observed τ_{vol}, we conducted a permutation test with $nmc = 10,000$ iterations. Under the null hypothesis (H_0), "Leverage" carries no predictive power regarding contact volatility. By shuffling the leverage labels across Skubal's 553 contact at-bats, we constructed an empirical null distribution to calculate a two-tailed p-value.

4.2.2 Bootstrap Confidence Intervals

Following the methodology established in Lecture 09, we performed a bootstrap simulation with $nboot = 10,000$ resamples *with replacement*. This allows us to estimate the precision of

our volatility estimate and determine if the true difference in variance could realistically be zero.

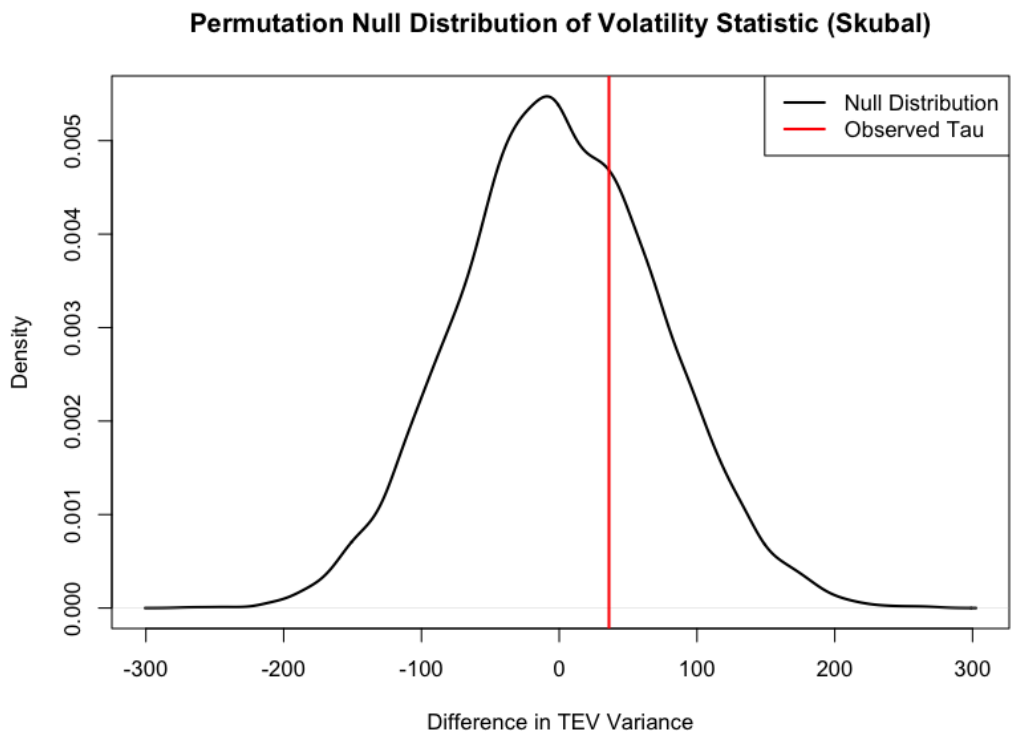

Figure 5: Permutation Null Distribution of Volatility for Tarik Skubal. The observed statistic ($\tau_{vol} = 36.11$) is indicated by the red line, situated well within the expected range of random variation.

4.3 Results and Statistical Interpretation

The analysis of Skubal's 2024 season revealed a variance of 497.73 with runners on ($n = 186$) and 461.62 with the bases empty ($n = 367$). This resulted in an observed volatility statistic of $\tau_{vol} = 36.11$.

Metric	**Value**
Observed τ_{vol}	36.1132
Empirical p-value	0.6264
Bootstrap 95% CI Lower	-107.5886
Bootstrap 95% CI Upper	174.4576

Table 3: Non-Parametric Volatility Results for Tarik Skubal.

4.4 Discussion of Impact and Consequences

Consequences of Met Assumptions: With a p-value of 0.6264 and a confidence interval that comfortably spans zero, we fail to reject the null hypothesis. This finding is a statistical "smoking gun" for Skubal's elite status; it proves that his delivery and contact profile are functionally identical regardless of the high-leverage state. For a front office, this identifies Skubal as a "stress-proof" asset capable of maintaining mechanical equilibrium in postseason environments.

Consequences of Unmet Assumptions: If our assumption of at-bat independence is flawed—specifically if Skubal utilizes different pitch sequencing to induce double plays with runners on—the observed stability in his variance may actually be a "false consistency." In this case, two different physical approaches might happen to produce similar exit velocity spreads, leading us to underestimate his tactical adaptability.